

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0624	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	P. Harthik	Reg. No.:	192524222
Section / Batch:		Date:	9/7/26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I P. Karthik - 192524222 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Karthik

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO1- AT 1

Name:- P. Karthek

Reg No:- 192524222

Code:- CSA0624

course:- DAA

① online video streaming platform
Given,

$$T(n) = 2T(n/2) + n$$

step 1: I identify Master Theorem parameter

$$a = 2$$

$$b = 2$$

$$f(n) = n$$

step 2: compute

$$n \log_b^a = n \log_2^2 = n$$

$$f(n) = \Theta(n \log_b^a)$$

This is master Theorem case 2.

Time complexity

$$T(n) = \Theta(n \log n)$$

Scalability Analysis

performance grows almost linearly with an additional logarithmic factor. suitable for handling millions of users efficiently.

Asymptotic Notation

$$\boxed{\Theta(n \log n)}$$

(2)

Social Media Feed processing
Given,

$$T(n) = T(n-1) + n$$

substitution method

Expand recurrence:

$$T(n) = T(n-1) + n$$

$$= T(n-2) + (n-1) + n$$

$$= T(n-3) + (n-2) + (n-1) + n$$

$$= T(1) + 2 + 3 + \dots + n$$

sum of first n numbers:

$$\frac{n(n+1)}{2}$$

Therefore,

$$T(n) = O(n^2)$$

performance analysis:-

- * Running time increase quadratically.
- * large inputs significantly reduce performance.

Asymptotic Notation:-

$$O(n^2)$$

③

e-commerce Recommendation system.

Algorithm 1

$$O(n \log n)$$

Algorithm 2

$$O(n^2)$$

comparison

Algorithm	Growth rate	Efficiency
$O(n \log n)$	slow growth	Better
$O(n^2)$	Fast growth	poor

Analyses:

- For small datasets, both perform similarly.
- For large datasets, $O(n \log n)$ is much faster

conclusion

$O(n \log n)$ is more scalable and efficient

④

cloud storage processing system

Given,

$$T(n) = 3T(n/2) + n$$

parameters

$$a=3, b=2, f(n)=n$$

compute

$$n \log_2^3 = n 1.585$$

$$n < n^{1.585}$$

$$T(n) = O(n^{1.585} - 2)$$

This is master Theorem case 1

Complexity

$$T(n) = O(n^{1.585})$$

Analyses

→ Better than Quadratic

→ suitable for large cloud storage system.

Financial Analytics system

Given,

$$T(n) = 2T(n/2) + n \log n$$

$$a=2, b=2, f(n) = n \log n$$

$$n \log_2^2 = n$$

$$T(n) = O(n \log n) = O(n \log_b^a \log_b^n)$$

This is Master Theorem Case 2
complexity

$$T(n) = O(n \log^2 n)$$

performance

- efficient for large financial datasets.
- slightly slower than $n \log n$.

$$(n \log^2 n) < (n \log n)$$